

Bank Customer Churn Analysis Report

1. Executive Summary

This project analyzes customer churn for a retail bank to identify key factors driving attrition and generate strategic recommendations to improve retention. Using customer demographic and account data, the analysis explores behavioral patterns of customers likely to exit the bank and highlights opportunities to reduce churn.

2. Objective

The primary objective of the analysis is to understand why customers are leaving the bank and identify customer segments at higher risk of churn, enabling the business to implement targeted retention strategies.

Data Sample:

customer_id	credit_score	country	gender	age	tenure	balance	products_number	credit_card	active_member	estimated_salary	churn	Age_Group	churn_out	Balance_Category	Credit_Score_Risk	tenure_group	products_category
15634602	619	France	Female	42	2	0	1	1	1	101348.88	1	Adult	Yes	No balance	Medium Risk	New Customer	Single product
15647311	608	Spain	Female	41	1	83807.86	1	0	1	112542.58	0	Adult	No	Medium	Medium Risk	New Customer	Single product
15619304	502	France	Female	42	8	159660.8	3	1	0	113931.57	1	Adult	Yes	High	High Risk	Long-Term	High Engagement
15701354	699	France	Female	39	1	0	2	0	0	93826.63	0	Adult	No	No balance	Low Risk	New Customer	Muti product
15737888	850	Spain	Female	43	2	125510.82	1	1	1	79084.1	0	Adult	No	High	Very Low Risk	New Customer	Single product
15574012	645	Spain	Male	44	8	113755.78	2	1	0	149756.71	1	Adult	Yes	Medium	Medium Risk	Long-Term	Muti product
15592531	822	France	Male	50	7	0	2	1	1	10062.8	0	Mature	No	No balance	Very Low Risk	Mid-Term	Muti product
15656148	376	Germany	Female	29	4	115046.74	4	1	0	119346.88	1	Young	Yes	Medium	High Risk	Mid-Term	High Engagement
15792365	501	France	Male	44	4	142051.07	2	0	1	74940.5	0	Adult	No	High	High Risk	Mid-Term	Muti product
15592389	684	France	Male	27	2	134603.88	1	1	1	71725.73	0	Young	No	High	Low Risk	New Customer	Single product
15767821	528	France	Male	31	6	102016.72	2	0	0	80181.12	0	Adult	No	Medium	High Risk	Mid-Term	Muti product
15737173	497	Spain	Male	24	3	0	2	1	0	76390.01	0	Young	No	No balance	High Risk	New Customer	Muti product
15632264	476	France	Female	34	10	0	2	1	0	26260.98	0	Adult	No	No balance	High Risk	Long-Term	Muti product
15691483	549	France	Female	25	5	0	2	0	0	190857.79	0	Young	No	No balance	High Risk	Mid-Term	Muti product
15600882	635	Spain	Female	35	7	0	2	1	1	65951.65	0	Adult	No	No balance	Medium Risk	Mid-Term	Muti product
15643966	616	Germany	Male	45	3	143129.41	2	0	1	64327.26	0	Adult	No	High	Medium Risk	New Customer	Muti product
15737452	653	Germany	Male	58	1	132602.88	1	1	0	5097.67	1	Mature	Yes	High	Medium Risk	New Customer	Single product

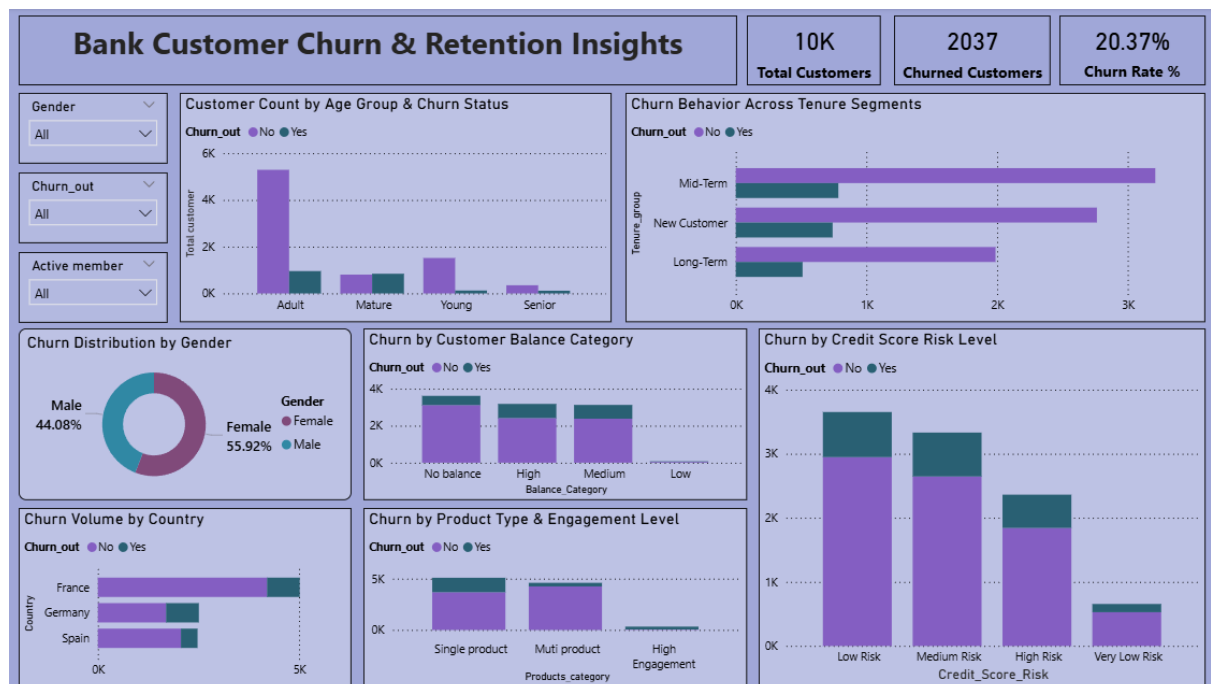
3. Data Overview

The dataset contains 10,000 customer records with the following key features:

- Customer demographics (Age, Gender, Country)
- Financial metrics (Balance, Estimated Salary, Credit Score)
- Account details (Tenure, Number of Products, Credit Card Ownership, Active Member status)

- Churn status indicating whether the customer exited the bank
- Feature engineering includes categories such as Age Group, Balance Category, Credit Score Risk, and Product Category

Dashboard:



4. Dashboard Insights

This dashboard helps us understand why customers are leaving the bank and what actions can be taken to reduce churn. The dataset contains 10,000 customers, and 2,037 of them have closed their accounts, resulting in a 20.37% churn rate.

What We Learned

- **Age:** Most customers who churned are Adults; Young and Senior groups show higher loyalty.
- **Gender:** Slightly more females churned than males, but the difference is minimal.

- **Tenure:** Mid-tenure customers leave more often; new and long-tenure customers tend to stay.
 - **Products:** Multi-product customers churn more, suggesting dissatisfaction or complexity.
 - **Balance:** Zero-balance customers show high churn; customers with consistent balances rarely exit.
 - **Credit Score:** Low and medium score groups churn the most; high-score customers are more stable.
 - **Country:** France has the highest churn, followed by Spain and Germany.
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5. Key Findings

Top insights identified:

- Higher churn is observed among customers with low tenure (new customers).
 - Customers with zero balance show an increased churn tendency.
 - Inactive members churn at a much higher rate compared to active customers.
 - High-risk credit-score customers have a greater likelihood of churn.
 - Certain geographic regions show higher churn than others (France, Spain, Germany comparison).
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6. Recommendations

Based on the analysis, the following actions are recommended:

- Improve engagement programs for customers in their first year.
- Provide financial incentives or product bundling for low-activity customers.
- Launch credit-improvement and financial-advisory support for high-risk credit-score customers.

- Improve communication and personalization to reduce churn in identified high-risk regions.
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7. Tools Used

- Power BI – Dashboard visualization & insights
- Python (Pandas, NumPy) – Data cleaning and exploration
- Excel – Initial dataset review

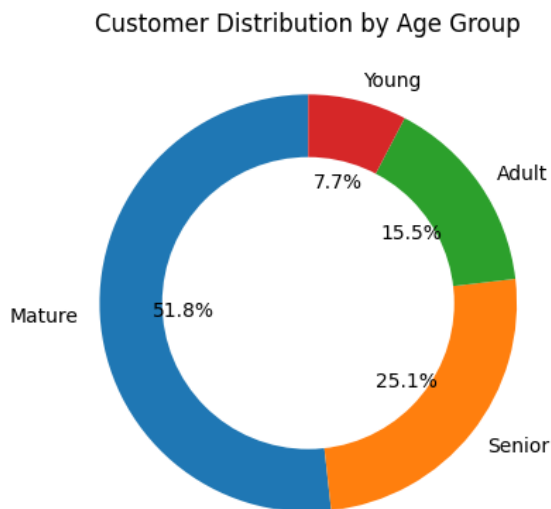
Summary:

	count	mean	std	min	25%	50%	75%	max
customer_id	10000.0	1.569094e+07	71936.186123	15565701.00	15628528.25	1.569074e+07	1.575323e+07	15815690.00
credit_score	10000.0	6.505288e+02	96.653299	350.00	584.00	6.520000e+02	7.180000e+02	850.00
age	10000.0	3.892180e+01	10.487806	18.00	32.00	3.700000e+01	4.400000e+01	92.00
tenure	10000.0	5.012800e+00	2.892174	0.00	3.00	5.000000e+00	7.000000e+00	10.00
balance	10000.0	7.648589e+04	62397.405202	0.00	0.00	9.719854e+04	1.276442e+05	250898.09
products_number	10000.0	1.530200e+00	0.581654	1.00	1.00	1.000000e+00	2.000000e+00	4.00
credit_card	10000.0	7.055000e-01	0.455840	0.00	0.00	1.000000e+00	1.000000e+00	1.00
active_member	10000.0	5.151000e-01	0.499797	0.00	0.00	1.000000e+00	1.000000e+00	1.00
estimated_salary	10000.0	1.000902e+05	57510.492818	11.58	51002.11	1.001939e+05	1.493882e+05	199992.48
churn	10000.0	2.037000e-01	0.402769	0.00	0.00	0.000000e+00	0.000000e+00	1.00

The dataset contains 10,000 bank customers, mostly around 39 years old with an average credit score of 650 and 5-year tenure. About 51% are active users, 70% have a credit card, and customers hold on average 1–2 products. The average balance is \$76K and estimated salary is about \$100K.

The churn rate is 20%, meaning about 1 in 5 customers left the bank.

Customer distribution by Age group:



Summary of Customer Distribution by Age Group

- Mature customers make up the majority at 51.8%
- Senior customers represent 25.1%
- Adult customers account for 15.5%
- Young customers are the smallest group at 7.7%

Insight: The customer base is dominated by mature and senior individuals, meaning banking services may be more aligned with older demographics rather than younger customers.

Conclusion:

Customer churn is heavily influenced by customer engagement, financial profile, and tenure. By applying segmentation and targeted retention strategies, the bank has a strong opportunity to reduce churn and improve customer lifetime value.